

Clinical Guideline for: **Diabetes in Pregnancy – Antenatal Care** Eastern Guideline

Summary

This guideline outlines the management of antenatal women who have diabetes and screening for gestational diabetes.

Key Points

- Refer URGENTLY to combined antenatal clinic or diabetes team as early as possible in pregnancy
- Ensure taking 5mg folic acid daily
- Aim to keep blood glucose <5.3 mmol/L before breakfast, < 6.4 mmol/L 2 hours after meals (adapted from NICE 2020)
- Recognise macrosomic babies as being particularly at risk of fetal demise
- Recognise that diabetic ketoacidosis has a high risk of fetal death
- Involve the regional maternal medicine network for complex cases if needed

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1. INTRODUCTION

This document gives guidance on the management of antenatal women with diabetes.

2. PRE-EXISTING DIABETES AND PREGNANCY

All women with Type 1 and Type 2 diabetes have an increased risk of **fetal malformation** (4-10%).

Women who also have hypertension, retinopathy or nephropathy are at high risk of pre-eclampsia and fetal growth restriction

All babies of mothers with diabetes have an increased risk of:

- Macrosomia.
- Fetal distress.
- Unexplained late stillbirth
- Neonatal hypoglycaemia
- Respiratory distress and other related problems caused by hyper-insulinaemia
- Intrapartum complications such as shoulder dystocia

3. PRE-CONCEPTUAL COUNSELLING

Optimal blood glucose control prior to conception reduces the risk of fetal malformation, miscarriage and other complications in pregnancy. However, the risk is reduced not eliminated. Pre-conceptual counselling has been shown to be cost effective (NICE 2008) and should be offered to all women with pre-existing diabetes.

The pre-pregnancy clinic for women living with diabetes takes place on a Tuesday afternoon in the diabetes centre and is led by the diabetes specialist nurses and dieticians. Women can self-refer to this clinic or be referred by their GP, other doctors or midwives.

Pre-conception discussion should include:

- The role of diet, healthy weight and activity levels
- The risks of hypoglycaemia and impaired awareness of it during pregnancy
- How nausea and vomiting in pregnancy can affect blood glucose control
- Increased risk of having a baby who is large for gestation which increases the chance of birth trauma, induction of labour and Caesarean section
- The need for retinal screening before and during pregnancy
- The need for diabetic nephropathy screening before pregnancy
- The importance of early feeding after birth to reduce the risk of hypoglycaemia
- The possibility that the baby needs to be admitted to the neonatal unit

- The risk of the baby developing obesity, diabetes and health conditions later in life.
- Use of reliable contraception until ready to conceive
- If conceives then advise to book with midwife before 8 weeks gestation
- Folic acid 5mg daily (prescribed dose) from 3 months prior to conception until 12 weeks' gestation to reduce the risk of fetal neural tube defects.
- Aim for HbA1c less than 48 mmol/mol if safely achievable but aim for the best possible HbA1c the woman can achieve. If the HbA1c is >86 mmol/mol, the woman should be **STRONGLY** advised NOT to become pregnant and to use reliable contraception (NICE 2020).
- Medication may need adjusting or altering prior to pregnancy e.g. stop statins, ACE inhibitors and angiotensin receptor blockers, and change anti-hypertensive.

4. COMBINED MEDICAL ANTENATAL CLINIC

- 4.1 Women with diabetes should be referred urgently to the diabetes team to arrange diabetes monitoring and seen at the next available combined antenatal clinic.

Please refer to the Diabetes Specialist Nurses via Epic by placing an order "DIABETES IN PREGNANCY" and follow all the prompts.

DSN telephone numbers are 01392 403823 and 403836.

The combined medical antenatal clinic is a multidisciplinary clinic for the management of pregnancy occurring in:

- Women with pre-existing Type 1 Diabetes
- Women with pre-existing Type 2 Diabetes
- Women with diabetes associated with cystic fibrosis
- Women found to have Diabetes in pregnancy (GDM – Gestational Diabetes)
- Women with other endocrine conditions requiring the opinion of the multidisciplinary team, e.g. thyroid problems

4.2 Staffing

- Consultant Endocrinologist (and Registrar in Endocrinology if available)
- Diabetes Specialist Nurse (DSN)
- Midwife with special responsibility for patients with gestational diabetes
- Midwife running the virtual thyroid follow-up service
- Consultant Obstetrician and Registrar in obstetrics
- Diabetes Specialist Dietician when available

4.3 In the Tuesday morning clinic our aims are

- To work as an effective, cohesive, multi-disciplinary team
- To build rapport with women who are pregnant and living with diabetes
- To support patients managing diabetes with courtesy and respect

- To acknowledge that managing diabetes in pregnancy can be very difficult
- To avoid being critical or undermining
- To empower women/couples to make choices based on carefully explained evidence
- To provide each patient with a clear explanation of what to expect from the clinic and the planned schedule of appointments throughout their pregnancy
- To refer to the regional maternal medicine network MDT for advice if required

5. MANAGEMENT OF PREGNANCY IN WOMEN WITH PRE-EXISTING DIABETES (TYPE 1 OR TYPE 2)

5.1 General Principles

- To discuss any co-existent medical or obstetric conditions, risks of pregnancy & plan management of the pregnancy
- To improve outcome of pregnancy by optimising diabetic control
- To manage complications of the diabetes
- To plan delivery and intra-partum management of diabetes
- Mothers with diabetes who attend for review or are admitted to hospital should be seen by a senior doctor at registrar or consultant level to check the care provided and formulate a discharge plan.
- Women with type 1 diabetes should be offered hybrid close loop (HCL) systems that are approved to be used in pregnancy and be provided with appropriate education and support to use these systems. The only pump licensed in pregnancy is the Mylife Ypsopump and the algorithm that accompanies this is cam APS FX (meeting standards of NICE / NHSE).
- Women with type 1 diabetes should be offered continuous glucose monitoring (CGM). Diabetes team to provide information on CGM and ensure women are educated in this form of monitoring. Education is given at the point of commencing, through the in-app tutorial and sharing platforms. Libre 2+ or Dexcom 1+ are most commonly used.
 - [Dexcom CGM and NHS funding - Type 1 Diabetes funding | Dexcom](#)
 - [Home | FreeStyle Libre | Abbott](#)
- The reviewing of the CGM data will then occur every 1-2 weeks either virtually or face to face at the diabetes multidisciplinary antenatal clinic appointments. Most women with diabetes will be reviewed every 4 weeks in the antenatal clinic, unless if closer follow-up is clinically required.
- Women with type 2 diabetes should have an objective record of their blood glucose recorded in their hospital records/EPR and be offered alternatives (e.g., CGM) to blood glucose monitoring if glycaemic targets are not achieved
- Women with diabetes should have an HbA1c measured at booking, and once during each trimester; those with an HbA1c above 48mmol/mol should be offered increased surveillance including additional diabetes nurse/dietetic

support, more frequent face to face review and input from their named, specialist Consultant to plan ongoing care and timing of birth decisions

- Women with complex diabetes, complex medical history retinopathy and/or kidney impairment requiring treatment during pregnancy can be discussed with the regional maternal medicine network (MMN). Can be referred to the weekly MMN MDT. However, as we have excellent senior input in Exeter both from a diabetes and renal point of view, patients do not all need to be seen in the regional unit. This has been agreed with the regional MMN.
- Women with type 2 diabetes on multiple doses of insulin (MDI) who decline continuous glucose monitoring should be encouraged to monitor at least 4 times per day before breakfast, 2 hours after each meal and before bed although more frequent monitoring may be necessary.

5.2 Referral

By GP/Community Midwife, Diabetes Specialist Nurse or Diabetes/Endocrine Consultant if pregnant to Consultant obstetricians or endocrinologists in the joint Obstetric/Diabetes & Endocrine clinic.

6. ANTENATAL VISITS

6.1 Blood Glucose Targets:

- 5.3 mmol/l before meals and <6.4 mmol/l 2 hours after meals

Most pregnant women with type 1 diabetes mellitus are using continuous glucose monitoring, the targets of their previous 14 days ambulatory glucose profile (AGP) should be:

- Pregnancy specific time in range (3.5-7.8mmol/L): ≥ 70%
- Time below range (<3.5mmol/L): <4% (Time below 3.0mmol/L:<1%)
- Time above range (>7.8mmol/L): <25%

6.2 Antenatal appointments for women with pre-existing diabetes

Holistic individualised care planning should be documented in the electronic patient record.

Pathways for additional identified needs should be initiated according to guidance e.g. psychological support, see *Women who present with mental health issues in the perinatal period* guideline.

Early appointment from 7 weeks if possible	Baseline HbA1c, U&Es, and thyroid function tests (in case of type 1 diabetes mellitus) and urine albumin creatinine ratio (UACR) (in case of all pre-existing diabetes mellitus) Retinal screening in each trimester (in the presence of retinopathy) or twice in pregnancy (if no retinopathy in the first trimester). BMI BP and urinalysis Check on folic acid 5mg Review oral hypoglycaemic agents Review medication
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	Advise Aspirin 150mg daily in the evening from 12 weeks Arrange a scan to assess viability Offer retinal screening if none in previous 3 months PCR if proteinuria
12 weeks	First Trimester screening with scan; obstetric and diabetes review Check commencing 150mg daily aspirin Document VTE assessment Folic acid can be stopped
16 weeks	Diabetes review in the ANC
20 weeks	Anomaly scan, obstetric and diabetes review Check HbA1c
24 weeks	Diabetes review in the ANC
28 weeks	Growth scan, obstetric and diabetes review Routine FBC, blood group and Anti-D if required. Retinal screening Document VTE assessment Check HbA1c. If >48 should be offered increased surveillance Document main issues/plan in Obstetric Management page (seen in labour by midwives)
32 weeks	Growth scan, obstetric and diabetes review Discuss plans for mode of delivery, including risks of shoulder dystocia. Book and consent for Caesarean if this is certain at this stage. Review/update Obstetric Management page
36 weeks	Growth scan, obstetric and diabetes review Advise to stop daily aspirin if they have had reasons to be on it Book induction of labour or Caesarean section Provide information about analgesia in labour and anaesthesia if needed Discussion pros and cons of steroids if planning a Caesarean Document plan for labour in Epic "Obstetric management" section Advice for managing diabetes during labour or Caesarean birth Advice for managing diabetes in the postnatal period Discuss breast feeding Provide kit with instructions for expressing and storing breast milk Document VTE assessment and plan for postnatal thromboprophylaxis Discuss options for contraception after delivery Encourage to continue with diet and regular exercise
37 -38+6 weeks	Pre-existing diabetes – deliver between 37 and 38+6 weeks as per NICE guidance. Please discuss timing of delivery with obstetric consultant.

- Urinalysis and BP recorded at each clinic visit
- Monthly PCR if pre-existing proteinuria.
- If proteinuria >500mg/daily for thromboprophylaxis (macroalbuminuria/ nephrotic syndrome).
- If proteinuria >200 mg daily or PCR >30 and creatinine >120umol/l, refer for renal opinion.
- More frequent visits with the diabetes team may be required if glycaemic control is poor. Many consultations now occur by telephone or e-mail.
- Steroids should be given if pre-term delivery is being planned but there is no evidence to support their use in diabetes after 36 weeks. May seriously increase glucose levels temporarily, consider admission if poor control.
- Routine fetal monitoring. e.g. CTG's, dopplers, not recommended before 38 weeks (NICE 2015).

6.3 Patients who decline planned delivery

Offer weekly fetal monitoring but be aware that this is not as reliable as in women who do not have diabetes. The risk of unexpected stillbirth is approximately 1:100

7. MANAGEMENT OF THE ILL PREGNANT WOMAN WITH DIABETES

7.1 Pregnant women with diabetes may become rapidly unwell with vomiting, diarrhoea, concurrent infection or any other significant illness.

Women who develop hyperglycaemia BG >13 or blood ketones >0.6 should be admitted urgently to exclude diabetic keto-acidosis as they are at risk of increased morbidity and perinatal mortality.

Diabetic ketoacidosis (DKA) is a serious condition and has a very high risk of fetal death. The obstetric consultant and diabetic consultant must be called immediately to review these cases in person, communicate effectively and have ongoing senior multidisciplinary input.

7.2 Treatment priorities on admission

1. Commence IV fluid replacement
2. Commence IV insulin infusion at a starting rate of 0.1units/kg/hour
3. Commence continuous cardiac monitoring and pulse oximetry
4. For the first 6 hours perform hourly capillary blood glucose and ketone measurements and 2 hourly U+Es via venous blood gas analysis. If capillary blood glucose meter reads "HI" perform a venous blood glucose.
5. Consider if patient requires referral to HDU / ITU (see guidance below)
6. Identify cause for DKA, including screening for infection and myocardial events.

See Clinical guideline for the management of Diabetic Ketoacidosis in adults.

8. OUTLIERS

Pregnant women with diabetes admitted to other wards on the hospital should be discussed on the labour ward handover and reviewed by a senior member of the team on a daily basis.

The plan for obstetric care should be summarised clearly on Epic and detailed on the Labour Ward board - phone Labour ward co-ordinator ext 6675.

9. SCREENING IN PREGNANCY FOR GESTATIONAL DIABETES

Based on NICE guidelines 2020 and HAPO study (2008)

9.1 Women with any one of these risk factors should be offered a glucose tolerance test at 26 weeks (24-28+6 weeks, NICE 2020) – see [appendix 1](#).

9.2 **Women who should be offered an oral glucose tolerance test**

- BMI over 30
- Previous large baby 4.5kg or above
- Previous gestational diabetes (see below)
- Family history of diabetes (first degree relative) Family origin with high prevalence of diabetes*:
 - South Asia (especially India, Pakistan and Bangladesh),
 - South East Asian
 - Middle Eastern (Saudi Arabia, Iraq, Syria, Kuwait, Jordan) Black African
 - Black Caribbean

Glycosuria 2+ on one occasion or 1+ on 2 occasions in first or second trimester

- Women taking steroids in pregnancy
- Previous unexplained stillbirth

(Maternal age > 40 and polycystic ovaries increase the risk but a GTT is not recommended for these reasons alone, there needs to be another risk factor in addition e.g. raised BMI).

9.3 Offer 75g glucose tolerance test at 24- 28+6 weeks

	Local Normal Values NICE 2020)
Fasting blood glucose	5.6 mmol/l or above
2 hours after 75g oral glucose	7.8 mmol/l or above

9.4 **Abnormal GTT (or high fasting glucose)**

Refer to the Diabetes in Pregnancy team via order on Epic and Tuesday morning antenatal clinic.

Measure HbA1c in pregnant women with abnormal OGTT. HbA1c $\geq$ 48mmo/mol suggests a pre-existing diabetes mellitus.

Dietary advice and glucose monitoring will be commenced as soon as possible and a clinic and growth scan appointment arranged.

9.5 Late diagnosis of GDM (from 29 weeks)

GTTs can be performed after 28 weeks if an earlier test was indicated but not performed for any reason. NICE guidelines no longer apply and abnormal results are more difficult to interpret. **There is no need to repeat a GTT if normal at the correct time for testing.** After 36 weeks the interpretation of an oral glucose tolerance test or capillary blood sugar monitoring is not possible to determine gestational diabetes and should NOT be used to make decisions about obstetric care. This means there may be some women where gestational diabetes cannot be excluded and decisions about timing of delivery and monitoring of neonatal blood sugar should be made by the responsible consultant in the light of this.

Please see [Appendix 2](#) re: guidance for late diagnosis of gestational diabetes.

9.6 Women with previous gestational diabetes or known impaired glucose tolerance

Book GTT as soon as possible in the pregnancy. Refer to DSN and ANC if this is abnormal. Repeat between 24 and 28+6 weeks if initially normal (NICE 2015). Commence glucose monitoring as an alternative, discuss with diabetes care team.

9.7 Women with a history of bariatric surgery

Caution: Currently there is no NICE or other National guideline for screening GDM in women post bariatric surgery. The following pragmatic recommendations are partly based on the consensus recommendations (see reference 6)

For pregnant women who have gastric band, at 24-28 weeks, offer OGTT as above, diagnose as GDM if fasting glucose is ≥ 5.6 mmol/L and/or ≥ 7.8 mmol/L 2h post OGTT

For pregnant women who have had bariatric surgery (except those who have gastric band), at 24-28 weeks, do not offer OGTT,

Arrange 7 days of CBG (capillary blood glucose) monitoring 4 times a day (before breakfast and 2h post meals).

- The cut off of fasting glucose ≥ 5.3 mmol/L
- The cut off of 2h post meal ≥ 6.4 mmol/L

Diagnosis as GDM: if any of the two fasting glucose values are at above the fasting cut off OR if any three (fasting and/or post meal) glucose values are at or above the cut offs.

10. MANAGEMENT OF MOTHERS WITH GESTATIONAL DIABETES

10.1 After diagnosis women should be contacted for:

- Capillary blood glucose monitoring: Can be taught either in the diabetes centre or ANC.
- Dietary advice: Can be arranged in the diabetes centre, ANC or over the telephone.
- The above is covered during a group education session which runs weekly. This is a face-to-face session run by a diabetes specialist nurse / midwife, & dietician
- An epic message is sent to the patient which includes details of the education appointment, eating well in diabetes leaflet, and a general information leaflet.
- [GDM Curriculum 2024 Update.docx](#)
- [Diabetes Team Contact Details General enquiries Email Rduh.docx](#)
- [Patient information leaflets | Intranet](#)
- To review treatment, which may be by diet alone, metformin or insulin depending on the stage of pregnancy and the blood sugars.
- In the antenatal clinic, a woman with GDM can be reviewed by the Diabetes Midwife and does not always need to be seen by an obstetrician.

10.2 Blood Glucose Targets

- < 5.3 mmol/l before meals
- < 6.4 mmol/l 2 hours after meals

10.3 Antenatal appointments for women with Gestational Diabetes

Holistic individualised care planning should be documented in the electronic patient record. Pathways for additional identified needs should be initiated according to guidance e.g. psychological support, see <i>Women who present with mental health issues in the perinatal period</i> guideline.	
12 weeks	GDM may be diagnosed early a GTT has been performed due to previous GDM.
16 weeks	Growth scans are not needed before 28 weeks
20 weeks	12 week FTSC and 20 week anomaly scan should be linked with Tuesday ANC A VTE assessment should be documented in each trimester
24 weeks	HbA1c to be checked at the first ANC visit
28 weeks	Growth scan, obstetric and diabetes review Routine FBC, blood group and Anti-D if required. Document VTE assessment
32 weeks	Growth scan, obstetric and diabetes review Discuss plans for mode of delivery, including risks of shoulder dystocia. Book and consent for Caesarean if this is certain at this stage.

36 weeks	Growth scan, obstetric and diabetes review Advise to stop daily aspirin if they have had reasons to be on it Book induction of labour or Caesarean section Provide information about analgesia in labour and anaesthesia if needed Discussion pros and cons of steroids if planning a Caesarean Advise stopping insulin when delivers Document plan for labour in Epic "Obstetric management" section Discuss breast feeding Provide kit with instructions for expressing and storing breast milk Document VTE assessment and plan for postnatal thromboprophylaxis Discuss options for contraception after delivery Encourage to continue with diet and regular exercise Advise early GTT in next pregnancy Advise HbA1c>13 weeks postnatal and annually thereafter Refer to the Diabetes Prevention Programme (with patient's consent)
38 weeks	Consider delivery if GDM and fetal macrosomia
39 - 40 weeks	Consider delivery if GDM with normal fetal growth and no complications (NICE says "to give birth" by T+6 days if all appears to be reassuring)

10.4 Steroids

Steroids to boost fetal lung maturity for women with diabetes should be used on an individual basis if a pregnant woman with diabetes needs preterm induction or elective Caesarean prior to 37 weeks' gestation. Please discuss with consultant in ANC.

11. REFERENCES

- 1) Diabetes in pregnancy: Management of diabetes and its complications from preconception to the postnatal period Clinical guideline [CG63] 2020
- 2) Management of diabetes in pregnancy 8, Management of diabetes, SIGN publication No 117, 2017 <http://www.sign.ac.uk>
- 3) Hyperglycaemia and Adverse outcomes in Pregnancy, HAPO study, 2008
- 4) IADPSG Recommendations on the Diagnosis and Classification of Hyperglycaemia in Pregnancy, Diabetes Care 2010 March 33(3) 676-682)
- 5) Battelino T, Danne T, Bergenstal RM, et al. Clinical targets for continuous glucose monitoring data interpretation: recommendations from the International Consensus on Time in Range. Diabetes Care 2019;42:1593–1603
- 6) Shawe J., Ceulemans D., Akhter Z., Neff K., Hart K., Heslehurst N., Štötl I., Agrawal S., Steegers-Theunissen R., Taheri S., et al. Pregnancy after bariatric surgery: Consensus recommendations for periconception, antenatal and postnatal care. Obes. Rev. 2019;20:1507–1522
- 7) <https://www.nice.org.uk/guidance/qs167/chapter/quality-statement-3-referring-people-at-high-risk-of-type-2-diabetes>
- 8) Diabetes Care . 2012 Jun 12;35(7):1492–1498. doi: [10.2337/dc11-2267](https://doi.org/10.2337/dc11-2267)
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3379591/>

12. MONITORING COMPLIANCE WITH THIS GUIDELINE

12.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, claims or complaints will be reviewed as per the Trust Policies regarding Incidents, Claims and Complaints, and may result in an audit and/or amendment to the guideline.

12.2 Relevant Policies:

- Incident reporting policy and procedure
- Claims management policy and procedure
- Policy and Procedure for the Management of Complaints, Concerns, Comments and Compliments

13. ASSOCIATED CLINICAL GUIDELINES OR POLICIES

Clinical guideline for the management of Diabetic Ketoacidosis in adults
Diabetes in pregnancy – Labour and the postnatal period

14. PUBLICATION DETAILS

Author of Clinical Guideline	Jenny Blackman, O&G Consultant Kash Patel, Bijay Vaidya, Kagabo Hirwa, Endocrinology consultants
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APPENDIX 1 – GUIDE TO PERFORMING A GTT

Guide to performing a Glucose Tolerance Test (GTT)

1. Call patient and check they have consumed nothing except water, have not smoked and have not chewed chewing gum since midnight.
2. Explain test – a blood sample will be taken and then the patient will be given Polycal, a carbohydrate drink, to consume within 10 mins. A second blood test will be taken 2 hours after completion of drinking the Polycal. In between blood tests the patient should sit quietly in the waiting area, should not smoke or chew gum but can drink water and go to the toilet if needed.
3. Take first blood sample in yellow topped bottle.
4. Give 113mls Polycal topped up to 200-250mls of water to taste.
5. Note time of completion of drink on blood bottle and blood form.
6. Give blood bag containing first sample and form to patient and move them to the waiting area. Explain they will be called by name for their second blood test in 2 hours and to notify a member of staff promptly if they have not been called.
7. Place patient label and time second test due on Phlebotomist's list.
8. Ensure patient label is in GTT book to ensure result is checked on the next working day.

APPENDIX 2 - Guidance for late diagnosis of gestational diabetes

(Professor Andrew Hattersley, Consultant Diabetologist RD&E NHS FT December 2019).
a.t.hattersley@exeter.ac.uk)

Following a recent multi-disciplinary team meeting, it has been agreed that the Diabetes Specialist Nurses will no longer accept referrals for capillary blood glucose monitoring in pregnancy, in order to diagnose gestational diabetes. If a clinical decision is made that a patient will need capillary glucose monitoring, this will need to be organised by the referring clinician (either via the midwife or practice nurse). It will also be the responsibility of the referring clinician to organise follow-up and review and interpretation of the blood glucose results. Please do not instruct the patient to email their results to the diabetic pregnancies email. We would suggest using the measurement of a laboratory fasting glucose which is a safe, convenient and accurate alternative. Following the measurement of fasting glucose referral to the diabetes nursing team for management would be based on the value of this (see below). The rationale for this decision are below.

Diagnosing gestational diabetes after 28 weeks using OGTT

There is no reference in NICE or in other guidelines as to how to interpret glucose values performed outside of 24-28 weeks. The reason that an early OGTT is suggested is that it gives sufficient time for education, monitoring and escalating to appropriate treatment.

Fasting glucose and post prandial glucose (1 or 2 hr post eating) in normal pregnancy increases by 10% from 28 weeks to 38 weeks. This is linear so approximately a 1% increase for every week after 28 weeks (Paretti et al Diabetes Care 2001 Aug; 24(8): 1319-1323.).
<https://www.ncbi.nlm.nih.gov/pubmed/11473063>

This would imply that interpreting an OGTT that is later than 28 weeks should have a higher cut off to be equivalent; at 32 weeks (4 weeks after 28 weeks) cut offs for gestational diabetes would be increased by 4% so fasting ≥ 5.8 mmol/l and 2hr > 8.1 .

Based on this, below are broad categories of diagnostic criteria based on gestational age:

	24-30 weeks	30-34 weeks	>34 weeks
Fasting	≥ 5.6 mmol/l	≥ 5.8 mmol/l	≥ 6.0 mmol/l
2hr	≥ 7.8 mmol/l	≥ 8.1 mmol/l	≥ 8.4 mmol/l

Capillary blood glucose monitoring is not recommended to diagnose gestational diabetes

The reasons to do glucose monitoring rather than an OGTT are very limited. This is only in situations where gestational diabetes is very likely that they will have ie previous gestational diabetes in previous pregnancy and unable to do an OGTT. In this situation it should only be implemented with dietary advice and then is being used as an indicator that additional treatment is needed over diet.

There is no evidence for the use of capillary glucose monitoring for the diagnosis of gestational diabetes either in guidelines or publications.

Capillary glucose monitoring is recommended in NICE as a measure of whether an increase in treatment is required. They specify that it should be used to indicate starting metformin if after *1-2 weeks of diet and exercise*, levels are not below 5.3 mmol/l fasting or 6.4 mmol/l 2hr post meal. **This would mean that doing glucose monitoring without advice on diet and exercise is not helpful for diagnosis or treatment.**

The diabetes nurses should not be required to teach glucose monitoring because of a missed OGTT or to interpret results to say if a diagnosis of GDM can be made.

Perform a fasting glucose in patients who refuse OGTT or who do not perform properly or have been missed

Late testing should only be considered when patient meets the criteria set out by NICE for testing for GDM. We suggest in this circumstance arranging the measuring of a laboratory tested fasting glucose value from blood collected in a fluoride tube (grey). This avoids the need for the patient to take the high sugar load needed in an OGTT and is easy to organise. It also means there are clear cut offs. We would suggest the gestation specific fasting cut offs proposed above from the OGTT but it could be that the 24-28 week fasting level should lead to further consideration. In the HAPO study the best value from OGTT at 24-28 weeks for predicting the risk of macrosomia was the fasting glucose.

From NICE they suggest fasting glucose in the OGTT as a very good way of detecting patients likely to have GDM that requires insulin. If doing this test, the priority is to not miss the most severe cases.

Decisions about possible early delivery should be made on obstetric grounds. If there remain doubt about whether the patient has gestational diabetes close to delivery and newborn glucose screening can be requested to detect possible fetal hypoglycaemia.